RUDRAKSH NALBALWAR

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EXPERIENCE

Software Engineer Intern — Feb 2025 – Apr 2025

Soroban Labs — *Nagpur, Maharashtra*

- Designed and implemented FastAPI-based backend APIs, integrating **LLMs for automated code generation** and video generation using the **Manim Library**, ensuring stability and maintainability.
- Developed automated video generation workflows, reducing manual coding time by approximately **70%** with an average API response time under 5 seconds.
- Created and maintained Bash-based deployment scripts, automating containerized deployments via Docker.

PROJECTS

FinGrounded RAG – Hybrid Financial Retrieval Platform

GitHub

Engineered a hybrid retrieval system combining **pgvector semantic search** and Postgres full-text search with **Reciprocal Rank Fusion**, achieving sub-100ms end-to-end retrieval via HNSW and GIN indexing. Built a SEC 10-K ingestion pipeline processing 100+ documents into 10K+ embedded chunks, and a fail-closed grounding validator ensuring 100% of answers are backed by verbatim evidence. Developed an async FastAPI backend and a React/TypeScript frontend with real-time streaming citations via Server-Sent Events.

Neo-Jarvis AI Assistant

GitHub

Building an agentic AI voice assistant in Python with a modular architecture, integrating LLM-powered conversational intelligence, autonomous web and desktop automation, real-time information retrieval, AI-driven content generation, and live OCR via webcam. Enabled multitasking through agent orchestration and concurrent execution using multi-threading.

Voice Feedback Agent

GitHub

Developed a Voice AI feedback agent using LiveKit, Deepgram STT, OpenAI GPT-4, and ElevenLabs TTS to enable natural Hinglish conversations and automate collection of structured feedback. Implemented real-time speech recognition, conversational flow management, and local transcript/JSON storage.

Heart Disease Predictor

GitHub — Live Demo

Developed a machine learning model using Python and Pandas to analyze diverse health datasets, achieving **92.5% accuracy** through hyperparameter tuning across Logistic Regression, KNN, Random Forest, and SVM. Deployed as a Flask-based web application for real-time heart disease risk predictions.

SKILLS

Languages	Python, Java, SQL, JavaScript, TypeScript, Bash
Backend & APIs	FastAPI, REST APIs, Node.js, OAuth2, JWT/Token Auth, async/await
Frontend	React.js, HTML, CSS, Tailwind CSS, Vite, shadcn/ui
Databases & Search	PostgreSQL, MySQL, pgvector, HNSW/GIN Indexing, Hybrid Retrieval (RRF)
AI/ML & Agents	LLMs, LangChain, LangGraph, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
Cloud & DevOps	AWS, Docker, Linux, Git, CI/CD, Supabase, Railway
Testing	Pytest, Unit & Integration Testing

EDUCATION

Bachelor of Technology in Artificial Intelligence and Machine Learning — Nov 2022 - Jun 2026

Shri Ramdeobaba College of Engineering and Management, Nagpur — CGPA: 7.3

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Deep Learning, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming

Higher Secondary Education — July 2022

Netaji Subhash Chandra Bose Jr. College, Nanded

ACHIEVEMENTS

- **Technical Leadership:** Led a team as Technical Head at NSS RCOEM to design, develop, and successfully launch the organization's first official website.
- Co-inventor of a granted design patent: Smart Desk Lamp with Automated Lighting and Wireless Charging (2025)Link
- Achieved 5-star Problem Solving rating on HackerRank.

CERTIFICATIONS

NVIDIA: Fundamentals of Deep Learning with PyTorch (Feb 2025)

FreeCodeCamp: Scientific Computing with Python (Dec 2023)